



Algorithmic Problem Solving

Practical Skills Module 4: Topic 3

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Topic Code	PM-04-PS03
Topic	Solve problems using algorithms

Scope of Practical Skill

Given a range the learner must be able to:

- PA0301 Solve problems using algorithms:
 - Dynamic programming algorithms
 - Searching and sorting algorithms
 - Mathematical algorithms
 - Data structures algorithms
 - Flow charts (on paper)

Applied Knowledge

- AK0301 Concept, definition and functions of intermediate Python programming principles

Internal Assessment Criteria

- IAC0301 Expected results with Python algorithms are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-04



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: Optimizing Delivery Routes for a Logistics Company

A logistics company needs to design a Python program to optimize delivery routes, minimize costs, and improve efficiency. The program will use algorithms and data structures to handle tasks like finding the shortest delivery route, managing delivery schedules, sorting packages by priority, and solving mathematical problems for cost and distance calculations. Flowcharts will be used to design and visualize these algorithms before implementation.

Create a Python file called **algorithms.py** and complete the following steps:

Step 1: Solve Problems Using Algorithms

Dynamic Programming Algorithm

Scenario: Optimize delivery routes to reduce total distance or cost. A common approach is the **Travelling Salesman Problem (TSP)**.

Instructions:

- Describe the steps to solve this problem in your own words or in pseudocode as comments inside the Python file you have created.
- Use a simplified example with 4–5 delivery locations.
- Explain how the algorithm finds the most efficient route that starts and ends at the same point.

Searching and Sorting Algorithms

Scenario: Packages must be sorted by delivery priority and searchable by ID.

Instructions:

- Implement a sorting algorithm to organise packages by priority level (e.g., "High", "Medium", "Low").
- Implement a searching algorithm to find a package by ID from a list.

Expected input example:

```
Package ID: P004, Priority: High  
Package ID: P002, Priority: Low
```

Expected output (after sorting):

```
P004 (High), P002 (Low)
```

Mathematical Algorithms

Scenario: Calculate delivery time and cost using distance, speed, and fuel rate.

Instructions:

Use the following formulas:

- $\text{time} = \text{distance} / \text{speed}$
- $\text{cost} = \text{distance} * \text{fuel_rate}$

Example input:

```
Distance: 120 km  
Speed: 60 km/h  
Fuel rate: R4.50/km
```

Expected output:

```
Estimated delivery time: 2.0 hours
Estimated fuel cost: R540.00
```

Data Structures Algorithms

Use a Stack to Track Packages:

- Add packages to a stack.
- Remove (pop) the most recent package.

Use a Queue for Delivery Scheduling:

- Add packages in delivery order.
- Remove (dequeue) packages when delivered.

Example output after operations:

```
Stack (top to bottom): P005, P004
Queue (front to back): P001, P002, P003
```

Step 2: Create a Flowchart

Purpose:

Visualise part of your delivery algorithm to plan it before coding.

Instructions:

- Choose **one** of the following processes to visualise:
 - Package sorting
 - Delivery scheduling
 - Cost calculation

Minimum steps your flowchart should show:

1. **Start**
2. **Input** data (e.g., package list or distances)
3. **Process**: Sort, assign, or calculate
4. **Decision** (optional): e.g., "Is priority High?"
5. **Output** result (e.g., delivery order or cost)
6. **End**

Draw your flowchart using:

- Ovals for Start/End
- Parallelograms for Input/Output
- Rectangles for Process steps
- Diamonds for Decisions
- Arrows to show flow

Submit your flowchart on paper or as an image.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
